Master's in Informatics and Computing Engineering

Department of Informatics Engineering

Dissertation Preparation

Scientific writing: from argument to sentence

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BEng

How much do I know about doing something?

Technical contributions

MSc

How much do I know about this specific subject matter?

Technical contributions

Scientific contributions

PhD

What new knowledge does the community gain from my contribution?

Technical contributions

Scientific contributions

Knowledge contributions

A dissertation is an argument.

Everything else follows from that.

Agenda

Argument

What you are claiming and what would make it credible

Document

the shape the argument takes as an object

Document Chapters

how each part opens, closes, and stays consistent

Paragraph

the unit in which an argument is actually made

Sentence

precision, tone, and citation

Non-textual elements

where all five levels meet on one object

A note on the examples

Throughout this class, you will see paired examples, one preferred and one to avoid.

All excerpts are illustrative.

They are written to isolate a single problem. Real writing fails in several ways at once.

Any citation appearing inside an excerpt is invented and should never be reused.

Argument

What you are claiming, and what would make it credible

Claim, evidence, warrant

Claim

What you assert to be true

Evidence

the measurement, experiment, proof or observation you offer

Warrant

Why does that evidence support a claim (and not some other explanation)

The warrant is what a student most often omits.

Numbers do not speak for themselves. A reader who cannot reconstruct your reasoning will supply their own — usually a less favourable one.

Claim, evidence, warrant in practice

Prefer

System A completed the workload 3x faster than System B.

The gain persists when the cache is disabled and across all three workloads, so it follows from the scheduling strategy rather than from memory locality.

Claim, evidence, and the reason they connect

Avoid

System A completed the workload 3x faster than B, which demonstrates the superiority of our scheduling strategy.

The warrant is missing! Nothing rules out caching, hardware, or a favourable workload

and "demonstrates" overstates what one comparison can show

Calibrating your claims

Match the strength of your verb to the strength of your evidence.

"suggests"

one experiment, limited conditions

"indicates"

consistent across several conditions

"demonstrates"

systematic evaluation, alternatives ruled out

"proves"

formal proof only — in CS, this word has a precise meaning

An overclaim is the easiest thing for a jury to attack and the cheapest to avoid.

What type of contribution are you providing?

Naming the type of contribution tells the reader what evidence would count as support.

a procedure or technique

a qualitative or descriptive model

an empirical model

an analytic model

a notation or tool

a specific solution or judgement

State it explicitly, early.
A reader who cannot tell what you are contributing cannot tell whether you succeeded.

What kind of validation are you offering?

Different contributions demand different validation. Say which you are giving.

Analysis — a formal or analytical argument

Evaluation — systematic measurement against alternatives

Experience — deployment, use in a real setting

Example — a worked case, a proof of concept

Persuasion — an argument that the approach is reasonable

A worked example is legitimate validation for some contribution types and inadequate for others.

Being explicit prevents the reader from applying a standard you never intended to meet.

Specify the context of your validation

An artefact (your program or system) is designed for a context.

Ask and answer in the text:

Under what conditions does the artefact work?

What did you assume about the context, and does it hold?

What happens when those assumptions fail?

State the limits of your evidence

This is a writing obligation, not a study-design topic.

Whatever your evidence is, it is bounded.

Define the bounds. How far does it reach?

Say what would be needed to remove these bounds.

A limitation you name is a limitation you control.

Raising the objection yourself and bounding it is stronger than leaving it for the examiner to raise.

Synthesising rather than listing

Organise by concept, not by author.

Author-centric

One paragraph per paper, in the order you happened to read them

Concept-centric

One section per idea, with the relevant sources arguing against each other

Build a concept matrix: sources down one axis, concepts across the other. The empty cells are your gaps.

The gap you identify is the foundation of your contribution. It must be visible in the review, not asserted afterwards.

Synthesising rather than listing

Silva et al. propose a caching layer. Costa et al. propose a prefetching strategy. Almeida et al. propose a hybrid method.

Author centric

Three summaries in sequence (good but not enough)

Approaches from Silva et al., Costa et al. and Almeida et al. assume a static workload. Yet none report behaviour under bursty traffic.

Concept centric

Compares, evaluates, and connects to your own work. The reader learns something that the sources do not say individually

The golden thread

Research question → Problem → Contribution → Evaluation → Conclusion

These five must be closely related to each other.

Every RQ is answered explicitly in the conclusions

Every claimed contribution is evaluated somewhere

Every evaluation traces back to a research question

Nothing is concluded that was not measured

If a link breaks, the jury will find it.

Document

The shape the argument takes as an artefact

Types of scientific manuscripts

Original research articles

Review articles

Short communications/letters

Technical reports

Case studies

Editorial/opinion

Books

Monographs (a long article or short book on a particular subject)

What a dissertation is….

A dissertation is a monograph and is an extended argument

It demonstrates logical, structured and defensible reasoning, based on credible evidence.

It is not:

a literature artefact (narrative, descriptive)

a report of activity - what you did, week by week

a project report – requirements, specification, development…

a tutorial on the technologies you used (procedural, instructional)

a description of a system you built

A system you built may produce evidence for a claim. It is not the claim.

Genre: argumentative or persuasive. You are defending a position.

A Master’s dissertation is the exercise of demonstrating

autonomous knowledge about a subject

The thesis as an object

The argument determines the document. The document is how a reader reaches the argument.

Front matter

Everything before Chapter 1

Body

The chapters that carry the argument

Back matter

references, then appendices.

Nothing load-bearing goes in an appendix. If the argument needs it, it belongs in the body.

Front matter

Abstract

context, problem, goals, approach, main results, main findings

written last; the most-read and least-revised part of most theses

Keywords

How your work will be found - choose terms others search for

Table of contents, list of figures, list of tables

generated, never typed

List of acronyms

Every acronym is still defined at first use in the text

Declarations

declaration of honour, and any other required statements

Body matter

A typical dissertation in CS, though your supervisor may vary it:

1. Introduction - context, problem, questions, contributions, methods, structure (optional background chapter)
2. Literature review and state of the art - what is known, and what is missing
3. Approach - the design, and why it is the right one
4. Implementation - what you built, at the level of detail a reader needs
5. Results - data collection from the implementation of your designed artefact
6. Evaluation - the evidence, and the limits of that evidence
7. Conclusions - what is now known, and what comes next

Chapter numbering is not a schedule. Chapters 3 to 7 mature as the work does.

End matter

What you write in the 1st semester and later

This semester

Introduction — context, problem, research questions

Literature review and state of the art — the review, the synthesis, the gap, the most recent related work

Approach — the proposed design, experiment and work plan

Next semester

Implementation — as it is built

Results – the data collection you achieve from running the experiment

Evaluation — data analysis as results arrive

Conclusions, contributions, abstract, document structure — last

You are learning the whole anatomy now so that you can write into a shape you already understand.

Term selection

Carefully choose a term to define each concept and never vary it.

Do not alternate user, participant, subject for the same role.

Do not alternate approach, method, technique, strategy when referring to the same thing.

Variation is a virtue in literary prose and a defect in technical prose.

A reader who meets a new term looks for a new idea.

Define critical terms precisely, at first use, and then hold them.

Not even when changing chapters!

Acronyms

Acronyms are a small language you are asking the reader to learn.

Define at first use. Example "*The United Nations (UN) are an international…*"

Keep a list in the front matter.

Do not reuse an acronym for two things, even in different chapters

Keep subscript and typeface conventions constant throughout

Inconsistent notation reads as carelessness, and carelessness in notation invites doubt about the results.

Tense

Past

What you did and what happened — "we measured", "the system failed“

For instance, to describe and experiment run (it already happened)

Present

established facts — "TCP guarantees ordering"

What the document does — "Section 3 presents"

What your artefact does — "the scheduler assigns“

What still holds currently. Example, “*the experiment protocol is divided into several parts…*”

Choose per category and stay consistent.

Drifting tense within a paragraph is one of the most common and most visible faults

Cross reference and referencing

Cross-reference integrity is a correctness property of your document.

Never submit a Section ?? or a Figure ??

Every figure, table, equation and algorithm is referred to in the text

Refer to an element before it appears, not after

Never type a number by hand - always use a label and a reference

Chapter, section and appendix numbering must match the contents and the lists

LaTeX gives you all of this. Let it fail loudly rather than fixing numbers manually.

The FEUP Dissertation LaTeX template

The official FEUP dissertation template [8] — LaTeX, feupteses.sty, report class, 11pt A4

Use the M.EIC option: \usepackage[meic]{feupteses}

Three lifecycle stages built into this template

no option - preparation stage

juri - version for evaluation by the committee

final - version for final submission, after acceptance

It fixes your front matter for you

abstract · UN SDG · acknowledgements · declaration of honour · TOC · list of figures · list of tables · acronyms

Bibliography via biblatex (\addbibresource, \PrintBib); iso and backrefs options available

Portuguese option available (portugues)

Fighting the template costs time you do not have. Learn it in week one.

Chapter

How each part opens, closes, and stays consistent

The sandwich structure

Every chapter opens and closes. The reader is never dropped into the middle.

1. Introduction

2. State of the art

2.1 Introduction … 2.n Summary

3. Approach

3.1 Introduction … 3.n Summary

N. Conclusions

The same pattern holds one level down: a long section opens and closes, too.

This is not decoration. It is how a reader who skims still follows the argument.

Writing a chapter introduction

Prefer

This chapter examines how existing scheduling strategies handle bursty workloads. It is an important topic because our deployment is bursty, and no reviewed approach addresses this case. Section 2.1 surveys static schedulers, 2.2 adaptive ones, and 2.3 sets out the gap this dissertation addresses.

What, why, and what is coming

Avoid

In this chapter, the state of the art is presented.

The heading already said that

No justification to read on, and no map of what follows

Writing a chapter conclusion or summary

A summary is not a repetition. It is what the reader should now be able to accept.

What were the key findings of this chapter?

What are the main takeaways of this chapter?

How do findings and takeaways advance the argument of the dissertation?

What does the next chapter have to do?

Test: is a reader getting the argument by just reading chapter introductions and summaries?

If the answer is no, the chapters are not carrying the thread.

Chapter sizes

Too long compared to other chapters? Perhaps divide?

Too short compared to other chapters? Perhaps a section in another chapter?

Try to balance chapter size!

Paragraph

The unit in which an argument is made

The paragraph as a unit of argument

One paragraph, one idea.

A paragraph is a move! A single move in the argument.

It asserts something, supports it, and connects it to what follows.

A section is a sequence of moves. The sequence must itself be an argument.

Where the reader looks

Readers do not weigh every word equally. Position carries emphasis.

Topic position — the start of a sentence

The reader expects the familiar: what this sentence is about

Use it to connect backwards, to what they already know

Stress position — the end of a sentence

The reader expects the new and the important

Put what you want remembered there

Most unclear technical prose is not too complex. It puts the wrong material in the wrong position.

Topic sentences

Prefer

Caching dominates latency in our workload.

States a claim that the paragraph will support

A reader who reads only this sentence still learns something

Avoid

This section discusses caching.

Announces a subject, asserts nothing

The reader must read the whole paragraph to find the point

Cohesion in sequence of ideas: old before new

Prefer

The scheduler assigns tasks to idle cores. Those assignments are recorded in a shared queue, which the profiler later reads.

Each sentence starts from where the last one ended

The reader is carried forward

Avoid

The scheduler assigns tasks to idle cores. A shared queue that the profiler later reads records those assignments.

The new element arrives first; the link backwards is buried

The reader re-reads to find the connection

Signposting and transitions

Tell the reader the shape of the argument before you make it and remind them where they are.

At the chapter level

The introduction forecasts; the summary confirms

At the section level

A sentence of transition when the argument turns

At the paragraph level

Connectives that name the relation: *however*, *therefore*, *by contrast, furthermore*

A transition must name a real relation. "Moreover" between unrelated paragraphs fools nobody.

The outline test

Extract the first sentence of every paragraph in a chapter and in writing order.

Read the resulting list on its own.

- Does it still make an argument?
- Is anything repeated?
- Is anything missing between two consecutive steps?
- Is anything in the wrong order?

If the list does not argue, the chapter should hold.

This takes twenty minutes per chapter and finds problems that no amount of proofreading will.

Sentence

Precision, tone and citation

Academic style and tone

Objective

Claims rest on evidence, not on enthusiasm

Precise

"a 23% reduction“, not "a relatively small reduction"

Formal

No colloquialisms (informal word or phrase used in casual, everyday conversation)
No rhetorical questions (to create an effect rather than to get an answer)

Evidence-based

Every non-obvious assertion is supported or attributed

Formal does not mean ornate. Plain, direct sentences are the most formal thing you can write.

Concision and precision

Prefer

The cache reduced the mean latency by 23%.

We evaluated three schedulers.

This approach fails under a bursty load.

Every word carries weight

Avoid

It can be observed that the introduction of the cache had the effect of producing a rather significant reduction in terms of the mean latency.

An evaluation was performed on a total of three different schedulers.

Padding, hedging, and nouns where verbs belong

Concision and precision

Attribute precisely. The reader must always know whose claim they are reading.

yours, supported by your data

yours, reasoned but not yet tested

someone else's, cited

generally accepted, cited to a survey or a standard

An unattributed claim is read as yours

If it is not yours, that is plagiarism

If it is yours and untested, it is an overclaim

Point of view

Plural first person is accepted in computing: "we measured", "our approach".

It denotes the team behind the work, even for a single author

It is usually clearer than the passive alternative

Some supervisors and some venues prefer impersonal forms. Ask your supervisor.

Whichever you choose, be consistent!

Switching points of view between sections is distracting and reads as though the chapters were written by different authors.

Writing in a second language

You are being judged on the argument, not on native fluency. But errors cost you credibility you cannot afford to lose.

Frequent traps for Portuguese speakers

false friends: actually ≠ atualmente, eventually ≠ eventualmente

pretend ≠ pretender

realise ≠ realizar

assist ≠ assistir

"informations", "researches", "softwares" — these are never plural

"permit to do" should be "allow us to do“

"in the figure below" should be "in Figure 4"

Write short sentences. Long sentences in a second language multiply the chance of error.

Grammar and spelling

Set your spell-checker to one variant of English and keep it there.

British or American — choose, but do not mix

For instance: visualisation (EN-UK) and visualization (EN-US)

Frequent faults

subject-verb agreement across a long subject phrase

"its" (possessive) and "it's" (it is — and never use it anyway)

"which" and "that"; "affect" and "effect"; "compose" and "comprise"

dangling participles: "Running the benchmark, the latency dropped"

A spell-checker will not catch a correctly spelled wrong word. Read aloud.

Citation styles

IEEE — numeric

"as shown in Silva *et al.* cache technique [3]"; references numbered in order of first citation

compact; standard across most of engineering and computing

ACM — numeric or author-year, depending on the template

ACM Reference Format; used across ACM conferences and journals

APA — author-date

"(Silva et al., 2023)"; common in HCI, education, and the social sciences

Numeric styles save space.

Author-date styles help readers who already know the literature.

Long documents often favour the latter.

Citation mechanics

M.EIC convention: IEEE numeric. Talk to your supervisor.

Cite where the claim is made, not at the end of the paragraph.

Do not start a sentence with a citation.

"[3] presents an empirical study…"

"Silva et al. [3] present an empirical study…"

Test: delete the bracketed number. The sentence must still read correctly**.**

Cite the work, not the author's reputation. Cite the version you actually read.

Reference hygiene

Never cite a work you have not read and verified in the original.

Citing from an abstract, from another paper's description, or from an unverified list will eventually be caught, and one false reference discredits the rest.

Check every entry against the publisher's record: authors, title, venue, volume, pages, year, DOI.

Do not trust exported BibTeX. Reference managers propagate errors faithfully.

Keep author-name formatting consistent across the whole list.

Prefer indexed, peer-reviewed sources. Put anything else in a footnote with a URL and an access date.

Non-textual elements

Elements that exist over the entire document

Non-textual elements intersect the previous 5 levels

For the example of figures…

Argument

Does the figure earn its place? What claim does the figure support?

Document

Is the figure numbered and listed in the list of figures?

Chapter

Is the figure placed near its discussion, and referred to before it appears?

Paragraph

Is the figure introduced, and then interpreted?

Sentence

Is the figure's caption precise and self-contained?

Figures

Use a figure for shapes, diagrams, and photos.

Earn its place

A figure that supports no claim should not exist

If a sentence would do the same work, write the sentence

Make it readable

vector format for diagrams and plots; check legibility at print size

axis labels, units, and a scale - always

never colour as the only channel: it fails in greyscale and for colour-blind readers

Place it

top or bottom of the page. A figure mid-page strands text around it

No figure appears without being discussed in the text.

Tables

Use a table when the reader needs (more) exact values

Design

Align numbers on the decimal point; keep significant figures consistent

Horizontal rules preferred. Vertical rules may add clutter, not information

Units in the header, not repeated in every cell

Read it

Tell the reader what to notice: the largest value, the outlier, the trend

A large table with no interpretation transfers your work to the reader.

Captions

Prefer

Figure 4: Mean latency against request rate for the three schedulers. The adaptive scheduler degrades beyond 800 req/s, where the others remain stable.

Self-contained: readable without the body text

Says what to notice, not only what the object is

Some supervisors may like smaller captions…

Avoid

Figure 4: Results.

Figure 4: Latency graph.

A label, not a caption

Forces the reader back into the text to learn anything

but not that small.

Equations

An equation is part of a sentence. Punctuate it accordingly.

Number only the equations you refer to later

Define every symbol at first use, in the surrounding text

Introduce it with a single sentence of lead-in

Interpret it afterwards: what does it mean, and why does it matter?

Display equations are for things you discuss. Inline mathematics is for things you merely use.

Algorithms and pseudo-code

Choose one level of abstraction and hold it

- Half prose and half source code serve neither purpose
- Omit language-specific detail unless it carries the argument

Make it referable

- Number the lines if you discuss specific steps
- State inputs, outputs, and preconditions
- State complexity, where it is part of your claim

Where appropriate, have your code supported by a repository.

Revision

They are different activities, and the order matters.

First, structural revision

Does the argument hold?

Is anything missing, repeated, or misplaced?

Cut! A shorter dissertation that argues well beats a long one that does not

Second, language revision

Paragraphs, then sentences, then words

Polishing a paragraph you will later delete is wasted work. Fix the thread first.

Proofreading

The last pass, on the text you have already decided to keep.

Leave time between writing and proofreading

Read aloud; your ear catches what your eye skips

Read backwards, paragraph by paragraph, to break the narrative pull

Check the mechanical things separately: numbering, references, captions, acronyms

Ask someone else - you cannot proofread your own text properly!

Your supervisor and your jury are busy. What reaches them should already be good.

Anti-patterns — argument and document

Argument

The orphan question — asked in Chapter 1, never answered

The missing warrant — evidence presented, reasoning left to the reader

The overclaim — "proves" on the strength of one pilot

The unexamined baseline — compared against a strawman

The activity report — what you did, in place of what you argue

Document

The author-by-author review — summaries in sequence, no synthesis

The load-bearing appendix — the argument depends on material outside the body

The last-minute abstract — written at 3 a.m. on submission day

Anti-patterns — chapter to sentence

Chapter

The cold open — a chapter that begins mid-topic, with no introduction

The shifting term — user, participant, subject, all the same person

The broken reference — Section ??, or a figure cited before it exists

Paragraph

The label topic sentence — "This section discusses X"

The two-idea paragraph — two moves fused into one

Sentence

The padded claim — twenty words where six would do

The unread citation — cited from an abstract, or never opened

Non-textual

The silent figure — beautiful, numbered, and never discussed

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Dissertation Preparation

Questions?